

# Basic Data types in python

1. Integer
2. float
3. Complex
4. Boolean
5. String

# What are strings?

→ String is a collection of unicode characters.

Unicode chars - Capital alph. (A-Z)  
Small alph. (a-z)  
Digits (0-9)  
Special char (@, \$, % ...)

String is written inside single,  
double or tripple quotes (intreved  
Comman)

eg. 'hello world' - single quotes  
"hello world" - double quotes  
'''hello world''' - Tripple quote  
''''hello world'''' - Tripple quotes

name = "ganesh"  
print(name) → ganesh  
print(tuple(name)) → str

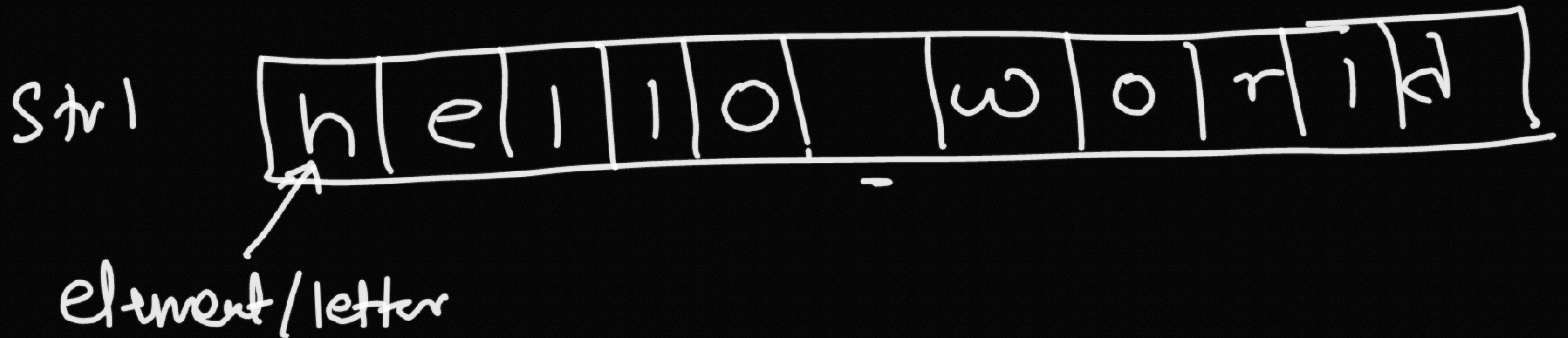
age = 25  
weight = 65.8  
eligible\_for\_vote = True  
num1 = 2 + 3j

To access the entire string or  
to display the entire string -  
we use that string variable  
to display it.

Str1 = "hello world"

print(Str1) → hello world

Indexing - Using indexing, we can access/display the individual elements of the String.  
String is a sequence of characters



Indexing - every element has position, called as index.

1) Positive indexing -

It starts from zero (first char)

2) negative indexing - Starts from  
-1 (last char)

# Positive Indexing

true  
indexing

0   1   2   3   4   5   6   7   8   9   10

str1

|   |   |   |   |   |  |   |   |   |   |   |
|---|---|---|---|---|--|---|---|---|---|---|
| h | e | l | l | o |  | w | o | r | l | d |
|---|---|---|---|---|--|---|---|---|---|---|

Print (str1) → hello world

Print (str1[0]) → h

Print (str1[1]) → e

Print (str1[2]) → l

Print (str1[3]) → l

Print (str1[4]) → o

Print (str1[5]) →

Print (str1[6]) → w

print (str1[7]) → o

print (str1[8]) → r

print (str1[9]) → l

print (str1[10]) → d

print (str1[11]) → IndexError

str2 =        0   1   2   3   4   5  
              m a h e s h

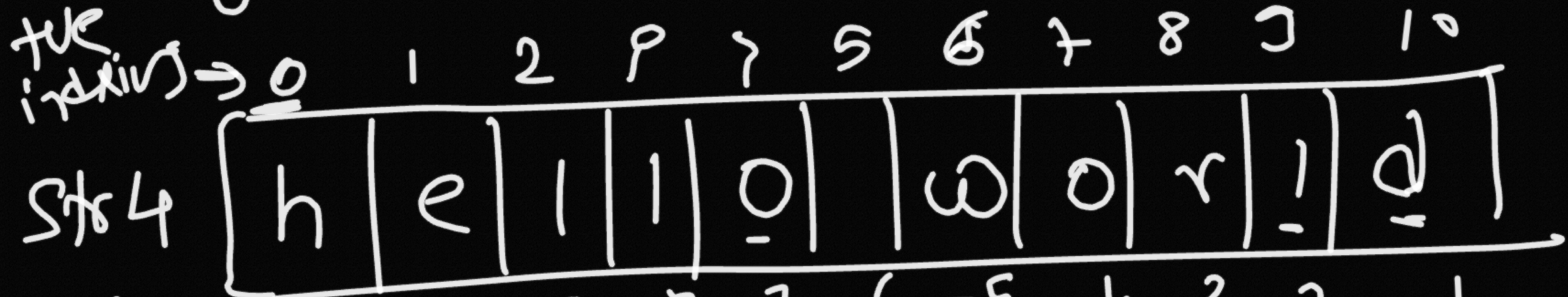
print(str2[0]) → m

print(str2[5]) → h

str3 = "wel come to python programming"  
         0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17  
         18 19 20 21 22 23 24 25 26 27 28

print(str3[8]) → t

# Negative indexing -



-ve indexing -11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1  $\leftarrow$

Print(str4)  $\rightarrow$  hello world    Print(str4[-4])  $\rightarrow$  o

Print(str4[-5])  $\rightarrow$  w

Print(str4[-1])  $\rightarrow$  d    Print(str4[-6])  $\rightarrow$

Print(str4[-2])  $\rightarrow$  !    Print(str4[-7])  $\rightarrow$  o

Print(str4[-3])  $\rightarrow$  r

print (str4 [-8]) → l

print (str4 [-9]) → l

print (str4 [-10]) → e

print (str4 [-11]) → h

print (str4 [-12]) → IndexError

Slicing - Using Slicing we can  
access substring (part of string)  
(group of elements)

# Slicing Syntax -

String\_name [Start : Stop<sup>+1</sup> : Step]

Start : start index for slicing. (default - 0)

Stop : stop index, slicing stops here  
stop index + 1

default  
len(str)

step : interval between characters  
(by default - 1)

|              |   |   |   |   |   |   |   |   |   |   |    |
|--------------|---|---|---|---|---|---|---|---|---|---|----|
| the indexing | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| Str6         | h | e | l | l | o |   | w | o | r | l | d  |

print(Str6) → hello world

print(Str6[0:3]) → hel

print(Str6[0]) → h

print(Str6[0:5]) → hello

print(Str6[6:11]) → world

print(Str6[6:8]) → wo



